

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 1

Due date: 12 September (Friday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let U be an upper triangular matrix. Suppose all the diagonal entries of U are zero. Show that there exists an integer N such that for all $k > N$, $U^k = 0$.

Sol: Let $u_{i,j}$ be the (i, j) entry of U and $U_{i,j}^k$ be the (i, j) entry of U^k . Since U is upper triangular and all the diagonal entries are zero, we know that $u_{i,j} = 0$ for $i \geq j$. When $k = 2$ and $i + 1 \geq j$,

$$U_{i,j}^2 = \sum_{t=1}^n u_{i,t} \cdot u_{t,j} = 0.$$

Similarly, if $U_{i,j}^k = 0$ for $i + k - 1 \geq j$, then for $i + k \geq j$

$$U_{i,j}^{k+1} = \sum_{t=1}^n U_{i,t}^k \cdot u_{t,j} = 0.$$

Hence $U^n = 0$.

2. Let K be the $n \times n$ matrix whose entries are 1 on the anti-diagonal entries and 0 elsewhere. The examples are: for $n = 2$,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

for $n = 3$,

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

for $n = 4$,

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Show that if U is a lower triangular $n \times n$ matrix, then KUK is an upper triangular matrix.

Sol: Let $u_{i,j}$ and $k_{i,j}$ be the (i, j) entry of U and K respectively, we know that

$u_{i,j} = 0$ for $i < j$, $k_{i,j} = 1$ if $i + j = n + 1$ and otherwise $k_{i,j} = 0$. Hence for $i + j > n + 1$,

$$(KU)_{i,j} = \sum_{t=1}^n k_{i,t} \cdot u_{t,j} = u_{n+1-i,j} = 0,$$

then for $i > j$,

$$(KUK)_{i,j} = \sum_{t=1}^n (KU)_{i,t} k_{t,j} = (KU)_{i,n+1-j} = 0,$$

hence KUK is upper triangular.

3. A matrix P is said to be a permutation matrix if for each row and column of P , there is precisely one non-zero entry in that row or column and such entry is one. Examples of permutation matrices include the identity matrix and the matrix K in Question 2. Other examples include

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Let A be an $n \times n$ matrix and let P be an $n \times n$ permutation matrix. Show that the number non-zero entries in A is the same as the number of non-zero entries in PAP .

Sol: Let $a_{i,j}$ and $p_{i,j}$ be the (i,j) entry of A and P respectively, assume $p_{1,x_1} = p_{2,x_2} = \dots = p_{n,x_n} = 1$ ($x_1, \dots, x_n$ are different numbers in $\{1, \dots, n\}$) and other entires of P are equal to 0. Then

$$(PA)_{i,j} = \sum_{t=1}^n p_{i,t} \cdot a_{t,j} = a_{x_i,j},$$

that means, the i -th row of PA is the x_i -th row of A . Similarly, if we assume $p_{y_1,1} = p_{y_2,2} = \dots = p_{y_n,n} = 1$, then the j -th column of AP is the y_j -th column of A . Therefor $(PAP)_{i,j} = a_{x_i,y_j}$, so the number of non-zero entries in A is the same as the number of non-zero entries in PAP .